

✓ CS407FinalTerm-70+MostImportantMCQswithAnswers

1. Which search algorithm always expands the node with the lowest path cost?

👉 A. Uniform Cost Search ✓

B. DFS

C. BFS

D. Hill Climbing

2. In A algorithm, $f(n) = ?$ *

👉 B. $f(n) = g(n) + h(n)$ ✓

A. $h(n) - g(n)$

C. $g(n)$

D. $h(n)$

3. Which of the following is a form of knowledge representation?

👉 D. All of the above ✓

A. Semantic network

B. Frames

C. Production rules

4. Which of the following search methods is complete and optimal?

👉 A. A* Search ✓

B. DFS

C. BFS

D. Hill Climbing

5. Which logic is used to represent uncertainty?

👉 A. Fuzzy Logic ✓

B. Propositional Logic

C. Predicate Logic

D. Modal Logic

6. Which component of an intelligent agent is responsible for taking actions?

👉 B. Actuators ✓

A. Sensors

C. Perceptron

D. Rules

7. Which technique is based on biological evolution?

👉 C. Genetic Algorithm ✓

A. Neural Network

B. Rule-Based System

D. Expert System

8. Which algorithm is based on gradient ascent?

👉 B. Hill Climbing ✓

A. BFS

C. DFS

D. GA

9. In depth-limited search, the depth limit is:

👉 A. Fixed ✓

B. Dynamic

C. Infinite

D. Random

10. What is the key issue in knowledge representation?

👉 D. All of the above ✓

A. Representing knowledge

B. Reasoning

C. Inference

11. Which AI domain deals with understanding human language?

👉 A. Natural Language Processing ✓

B. Robotics

C. Expert System

D. Machine Vision

12. Which AI agent operates in the present and does not think ahead?

👉 C. Reflex Agent ✓

A. Goal-based Agent

B. Utility-based Agent

D. Learning Agent

13. Which algorithm uses a heuristic function to guide the search?

👉 A. Greedy Search ✓

B. BFS

C. DFS

D. IDDFS

14. What does $h(n)$ represent in A algorithm?*

👉 B. Estimated cost to reach the goal ✓

A. Actual cost

C. Total cost

D. Path length

15. Which of these is not a type of learning?

👉 D. Passive Learning ✓

A. Supervised

B. Unsupervised

C. Reinforcement

16. Which of the following is not a component of PEAS?

👉 C. Solver ✓

A. Performance

B. Environment

D. Sensors

17. Which problem-solving approach explores all nodes equally?

👉 A. Breadth First Search ✓

B. Depth First

C. A*

D. Greedy

18. A knowledge base contains:

👉 B. Facts and Rules ✓

A. Questions

C. Heuristics

D. Goals

19. Which type of agent improves its performance with learning?

👉 D. Learning Agent ✓

A. Simple Reflex

B. Goal-based

C. Utility-based

20. In Prolog, facts and rules are written using:

👉 A. Predicate Logic ✓

B. Set Theory

C. First-Order Algebra

D. Probability

21. The goal of AI is to develop systems that can:

👉 C. Mimic human intelligence ✓

A. Store data

B. Run faster

D. Solve puzzles

22. Which of these uses backtracking?

👉 B. Depth First Search ✓

A. Breadth First

C. A*

D. Hill Climbing

23. Which of these is a drawback of hill climbing?

👉 D. All of the above ✓

A. Local Maxima

B. Plateaus

C. Ridges

24. Which search algorithm is suitable for infinite trees?

👉 C. Iterative Deepening ✓

A. BFS

B. DFS

D. A*

25. Which learning algorithm mimics the brain?

👉 A. Neural Networks ✓

B. Genetic Algorithm

C. Decision Trees

D. Q-Learning

26. The perceptron is a unit of:

👉 B. Neural Network ✓

A. Prolog

C. Semantic Network

D. Rule-Based System

27. Which of these is an uninformed search strategy?

👉 A. BFS ✓

- B. A*
- C. Greedy
- D. Best-First

28. Which is not a characteristic of intelligent agents?

-  C. Manual control ✓
- A. Reactivity
- B. Autonomy
- D. Proactiveness

29. Which search algorithm is space efficient?

-  B. DFS ✓
- A. BFS
- C. A*
- D. UCS

30. The function that evaluates how close a state is to the goal is called:

-  A. Heuristic ✓
- B. Depth
- C. Cost
- D. Utility

31. Which problem-solving agent performs well in a deterministic environment?

👉 C. Model-Based Agent ✓

A. Random Agent

B. Blind Agent

D. Passive Agent

32. Which search method is best when the solution is far from the root?

👉 B. DFS ✓

A. BFS

C. A*

D. Best First

33. Which of these terms is associated with Genetic Algorithms?

👉 D. Crossover ✓

A. Node

B. Goal

C. Arc

34. Which learning type works without labeled data?

👉 B. Unsupervised Learning ✓

A. Supervised

C. Reinforcement

D. Transfer Learning

35. Which algorithm uses a priority queue for implementation?

👉 A. A* ✓

B. DFS

C. BFS

D. Hill Climbing

36. Which is a reactive AI system?

👉 A. Reflex Agent ✓

B. Neural Net

C. Planner

D. Deep Agent

37. Which of these is NOT part of the expert system?

👉 C. Compiler ✓

A. Inference Engine

B. Knowledge Base

D. User Interface

38. Which search algorithm is guided only by heuristics?

👉 A. Greedy Best-First ✓

B. BFS

C. DFS

D. UCS

39. Which agent maintains internal state?

👉 C. Model-Based Agent ✓

A. Reflex Agent

B. Learning Agent

D. Simple Agent

40. Which of these works best in partially observable environments?

👉 D. Model-Based Agents ✓

A. Reflex

B. Goal-based

C. Utility-based

41. Which knowledge representation is used in semantic nets?

👉 B. Graphs ✓

A. Trees

C. Tables

D. Lists

42. Which learning technique uses rewards and punishments?

👉 A. Reinforcement Learning ✓

B. Supervised

- C. Transfer
- D. Batch

43. The concept of alpha-beta pruning is used in:

👉 B. Minimax algorithm ✓

- A. BFS
- C. DFS
- D. GA

44. What kind of task is classification in AI?

👉 A. Supervised Learning ✓

- B. Unsupervised
- C. Reinforcement
- D. Reactive

45. FOPL stands for:

👉 A. First Order Predicate Logic ✓

- B. First Objective Propositional Logic
- C. Forward Order Processing Logic
- D. None

46. In Prolog, a goal is satisfied when:

👉 C. It matches a fact or a rule ✓

- A. It is defined in a program
- B. It returns false
- D. It is found in a loop